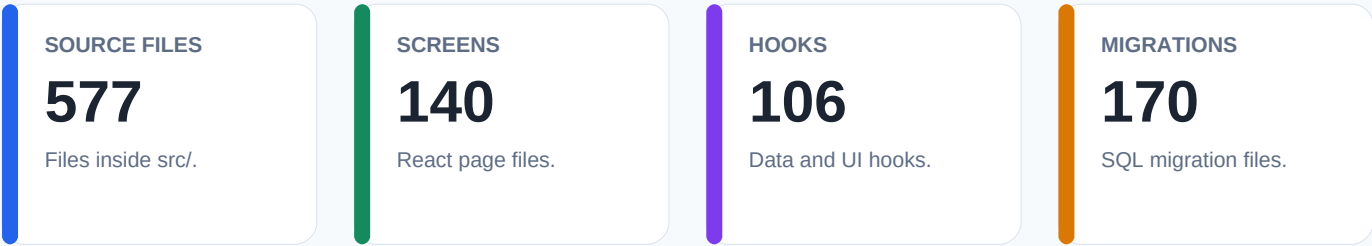


XO Platform

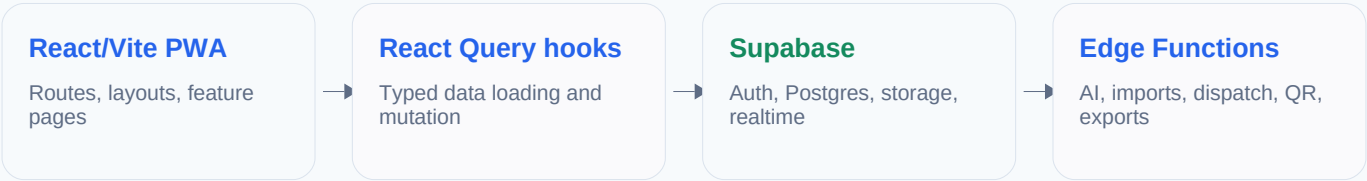
Feature Inventory, Graphic Statistics, and Clean Code Strategy

Generated from local repository on 2026-07-19



Executive Snapshot

This project is a multi-role nursery operations platform: React/Vite on the client, Supabase for Auth/Postgres/Storage/Realtime, and Supabase Edge Functions for operational workflows. The feature surface is broad and mature, but current build quality is blocked by TypeScript errors and database type drift.

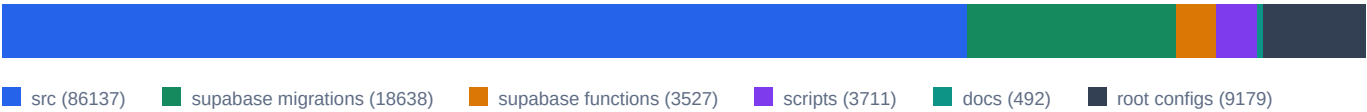


Current Verification Status

Check	Result	Meaning
npm run typecheck	FAIL	TypeScript build currently fails. Highest-impact issues: TeacherLayout missing names, stale Supabase database types, resolver/type mismatches.
npm run lint	FAIL	110 problems: 78 errors, 32 warnings. Main clusters: React Hooks purity/set-state-in-effect, Fast Refresh exports, unused values.
PDF generation	PASS	This report was generated locally with jsPDF from repository files and observed command output.

Graphic Statistics

Live counts from the repository excluding node_modules, dist, .git, and attached binary assets.



TOTAL LINES

121,684

Text/code lines counted locally.

COMPONENTS

180

Reusable TSX components.

ROUTES

147

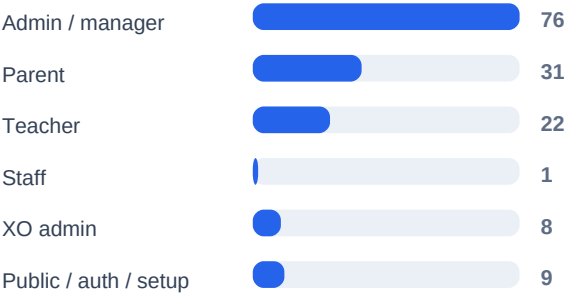
Declared app routes.

EDGE FUNCS

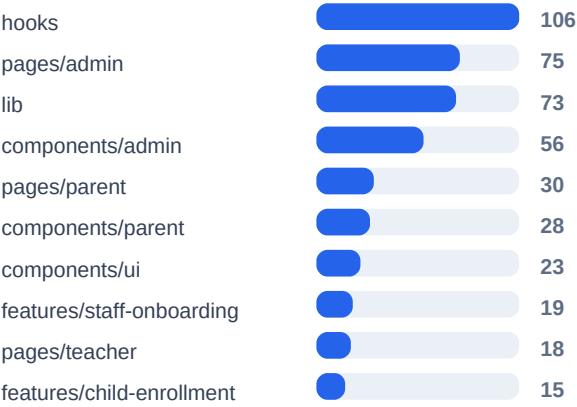
16

Supabase functions.

Routes by Audience



Source Areas by File Count



Repository Code Mix

Area	Files	Lines
Src	577	86,137
Supabase Migrations	170	18,638
Supabase Functions	20	3,527
Scripts	22	3,711
Docs	5	492
Root Configs	48	9,179

Feature Inventory

Grouped by product domain and primary audience. This is the readable map of the project surface.

Feature domain	Audience	What exists in the codebase
Platform admin	xo_super_admin	Nursery CRUD, subscription/pricing setup, platform analytics, XO payment overview.
Nursery operations	admin, manager	Dashboard, onboarding, classes, children, calendar, settings, role/position/feature admin.
Admissions and enrollment	admin, parent	Public inquiry, parent signup, applications, waitlist, import, child enrollment wizard.
Children and attendance	admin, teacher, parent	Child profiles, attendance dashboards, QR scanner, pickup verification, history.
Staff and HR	admin, manager, staff	Staff directory, onboarding, schedules, attendance month view, payroll, payslips.
Events and permissions	admin, parent, teacher	Event CRUD, parent event creation, permission responses, reminders, QR/event attendance, invoices.
Finance and billing	admin, parent, xo_admin	Invoices, payment attempts, Paymob initiation, financial dashboard/reports, packages, loyalty.
Reports and learning	teacher, parent, admin	Daily reports, feeding/nap/mood/toilet sections, milestones, quarterly reports, courses.
Media and content	admin, teacher, parent	Upload, compression, privacy/visibility, approval, gallery, downloads, content library.
Communication	all roles	Notifications, web push, chat, inbox/messages, broadcasts, email/SMS/WhatsApp dispatch.
Surveys and forms	admin, parent	Survey builder, renderer, responses, analytics panel, targeted notifications.
Health and safety	admin, parent, teacher	Allergies, conditions, medications, vaccinations, documents, alert dashboard, update requests.
AI and help	all roles	Help articles, route tracker, AI assistant, AI action safety, AI import mapper.
Backup and ops	admin, ops	Tenant export, usage watch, restore rehearsal, integrity and cross-tenant denial scripts.
PWA, i18n, theming	all roles	Service worker, offline indicator/queue, Arabic/English locale files, RTL notes, theme switching.

Role and Permission View

The current permission matrix is defined in src/lib/permissions/matrix.ts. XO super admin bypasses this matrix.

	Top mgmt	Manager	Teacher
Dashboard Attendance	full	full	full
Dashboard Finance	full	finance	none
Kids Applications	full	full	none
Staff	full	hr	none
Classes	full	full	full
Admissions	full	full	none
Event Calendar	full	full	full
Financial Reports	full	finance	none
Notifications	full	full	full
Media Library	full	full	approval
Upload Media	full	full	full
Content Library	full	full	full
Surveys	full	full	none
Messages	full	full	full
Daily Reports	full	full	approval
Child Enrollment	full	full	none
Staff Onboarding	full	full	none
Inventory	full	full	none
Meals	full	full	none
Qr Code	full	full	none
Health Alerts	full	full	full
Loyalty	full	finance	none
Broadcast Messages	full	full	none

Permission Model Notes

Top management has full access across the feature matrix. Managers are mostly full access, with finance-only and HR-only exceptions. Teachers have direct classroom/attendance/event/report/media surfaces, with some report/media flows marked as approval-based. Parents are handled outside this admin matrix through parent-child relationships and protected parent routes.

Backend and Data Shape

Supabase is the operational backbone: typed client, SQL migrations, Edge Functions, storage buckets, and role-aware RLS helpers.

DB TABLES

106

Unique CREATE TABLE names found.

SQL FUNCS

66

Unique CREATE FUNCTION names.

USED TABLES

81

Supabase .from(...) names in code.

TYPE DRIFT

28

Used names missing from database.ts.

Edge Function Map

Function	Likely responsibility
ai-assistant	Help/AI assistant endpoint
ai-import-mapper	AI-assisted import mapping
child-enrollment-complete	Finalize child enrollment workflow
email-dispatch	Email notification dispatch
emergency-broadcast	Urgent broadcast workflow
event-qr-issue	Issue QR tokens for events
parent-signup-complete	Finalize public parent signup
payment-reminders	Scheduled invoice/payment reminders
permission-deadline-reminder	Event permission deadline reminders
process-import	Spreadsheet/import job processor
qr-token	QR token issuance
qr-verify	QR verification and attendance/pickup handling
sms-dispatch	SMS dispatch
staff-onboarding-complete	Finalize staff onboarding workflow
tenant-export	Tenant-scoped backup/export
whatsapp-dispatch	WhatsApp dispatch

Database Type Drift - Highest Impact

Missing from src/types/database.ts	Why it matters
chat_conversation_members	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
chat_conversations	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
chat_messages	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
child_diaper_care	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
child_dietary_preferences	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
child_packages	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
course_enrollments	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
course_invoices	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
courses	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
event_attendance	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.

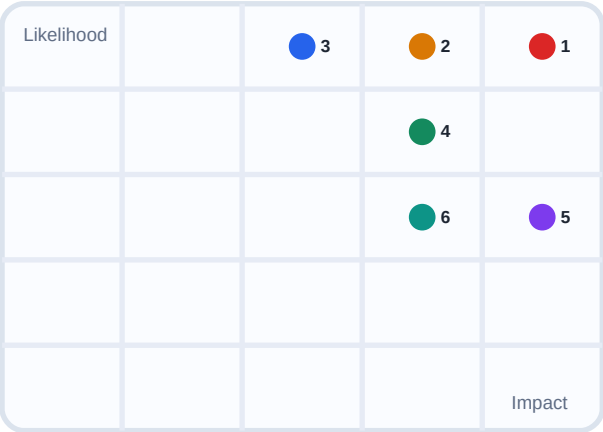
Missing from src/types/database.ts	Why it matters
event_attendees_public	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
features	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
import_jobs	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
packages	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
personal_reminders	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
pickup_incidents	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
positions	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.
quarterly_report_grade_levels	Code calls supabase.from(...) for this object, but TypeScript cannot infer rows/inserts/updates correctly.

Clean Code Findings

Prioritized issues and cleanup direction based on static inspection plus typecheck/lint output.

Priority	Finding	Recommended fix
P0	Build blocked	npm run typecheck fails. TeacherLayout has missing names; DB type drift creates many Supabase never-row errors.
P0	Teacher shell broken	TeacherLayout.tsx references useUnreadMessagesCount, teacherDisplayName, profile, and languagePref without valid definitions.
P0	Supabase types stale	Used backend objects missing from database.ts include: chat_conversation_members, chat_conversations, chat_messages, child_diaper_care, child_dietary_preferences, child_packages, course_enrollments, course_invoices, courses, event_attendance, event_attendees_public, features,
P1	Form type contracts	Staff onboarding and course forms have resolver/default-value mismatches. Use z.input/z.output or align form values with schema transforms.
P1	Lint signal is high	npm run lint reports 110 problems: 78 errors and 32 warnings. Main clusters: React Hooks purity, set-state-in-effect, fast-refresh exports, unused values.
P1	Schema migration drift	Many migrations add domain tables after the manual type file. Add automated Supabase type generation and a drift check in CI.
P2	Documentation gap	README is still the default Vite template. Add architecture, environment setup, role matrix, feature map, and operational runbooks as first-class docs.
P2	Feature sprawl risk	106 hook files and broad admin surfaces are workable, but future work should move toward feature-owned modules with local hooks, UI, schemas, and tests.

Risk Map



Legend

- 1. Build/type safety - Failing typecheck blocks reliable deploys.
- 2. Tenant isolation/RLS - Many migrations target recursion/security. Needs regression.
- 3. Payments/invoices - Financial flows exist across admin, parent, and XO admin.
- 4. Notifications/dispatch - Email/SMS/WhatsApp/push need retry and audit trails.
- 5. Feature growth - Lots of pages/hooks can become hard to reason about without ownership.
- 6. Documentation - Current README does not describe the real system.

Immediate Fix Order

Order	Action	Benefit
1	Fix TeacherLayout missing imports/derived values.	Restores an app shell and removes hard compile errors.
2	Regenerate or extend Supabase Database types for new tables/functions/views.	Clears the largest cluster of never-row TypeScript errors.
3	Align form schemas with useForm generics in staff onboarding and courses.	Keeps validation, defaults, and submit payloads consistent.
4	Decide React Compiler lint posture and fix repeated patterns deliberately.	Turns lint from noise into a dependable gate.
5	Add smoke tests for each role route group.	Prevents broken shells/routes from reaching users.

Best Structure Strategy

A practical architecture direction for making the codebase easier to understand, change, and verify.

Week 1
Restore build health

Fix TeacherLayout, regenerate Supabase types, correct resolver generics, remove unused imports.

Week 2
Stabilize quality gates

Make typecheck/lint required in CI; tune React Compiler lint policy; add smoke tests for route shells.

Weeks 3-4
Vertical slice cleanup

Move admin/parent/teacher features into domain folders with shared UI kept small and intentional.

Month 2
Database discipline

Canonical schema docs, generated types, migration review checklist, RLS regression tests.

Month 3
Operational maturity

Backup restore rehearsal, edge-function observability, bundle budgets, role-permission audit UI.

Strategy	How to apply it in this project
Use feature slices	Group page + hooks + domain lib + schema near each feature. Keep src/components/ui for primitive reusable controls only.
Generate database types	Replace hand-maintained Database tables with generated Supabase types; run drift check whenever migrations change.
Keep route shells thin	Layouts should compose navigation, profile/session queries, notification drawers, and Outlet only. Move business logic into hooks.
Define domain service modules	For each feature, isolate Supabase reads/writes in typed hooks or lib modules, not scattered across pages.
Add contract tests	Cover RBAC gates, RLS-sensitive data access, payment/event flows, and onboarding validation before large refactors.
Document ownership	Every feature gets owner, routes, tables, edge functions, and critical user flows in a short living doc.

Recommended Target Folder Shape

src/features/attendance/{pages,components,hooks,api,types}.ts
src/features/events/{pages,components,hooks,api,schemas}.ts
src/features/finance/{pages,components,hooks,api,formatters}.ts
src/components/ui/* for design-system primitives only
src/lib/supabase/* for generated database types and shared client helpers
tests/e2e/role-smoke.spec.ts and tests/rls/*.sql for regression coverage

Definition of Clean for Future Work

Gate	Expectation
Typecheck	npm run typecheck passes before merge. Generated DB types are current.
Lint	npm run lint has zero errors. Warnings are intentional and tracked.
Feature ownership	Every route has a clear feature owner and linked tables/functions.
Tests	High-risk flows have smoke or integration coverage: auth, RBAC, payments, event permissions, onboarding, imports.
Docs	README explains the real app; docs/ contains feature inventory, setup, backup, and RLS notes.

Appendix

Machine-readable-ish details used to make the graphics.

Package / stack item	Detected value
App name	xo-platform
Build tool	Vite + React + TypeScript
React	^19.2.4
Router	^7.13.2
Data fetching	^5.95.2
Backend client	^2.100.0
Charts in app	^3.8.1
PDF library in repo	^4.2.1 / autotable ^5.0.8
Internationalization	^25.10.9 / ^16.6.6

File Type Counts

Extension	Files
.tsx	344
.ts	254
.sql	174
[none]	15
.mjs	14
.json	12
.md	9
.svg	4
.js	3
.png	3
.pdf	2
.bak	1
.css	1
.example	1